



Maincrafts Technology | AI & ML Internship Program

Project Report **on** **“California House Price Predictor”**

Predictive Modeling using Linear Regression

Internship Task:

House Price Predictor Workflow - Task [1]

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1. Introduction

The objective of this project was to build a predictive model using **Linear Regression** to estimate median house values in California. This task follows the complete Machine Learning lifecycle, including data acquisition, exploratory data analysis (EDA), model training, and performance evaluation. This project serves as a foundational exercise in understanding regression analysis and feature significance within the context of the **Maincrafts Technology** internship curriculum.

2. Exploratory Data Analysis (EDA)

Before training the model, the dataset was analysed to understand the relationships between different features (such as average income, housing age, and location) and the target variable (Median House Value).

2.1 Data Quality Check

- **Missing Values:** The dataset was checked for null entries. It was found to be clean, with no missing values across the 20,640 samples.
- **Statistical Insights:** The average median house value is approximately 2.06 (in hundreds of thousands of dollars).

2.2 Feature Correlation

Correlation analysis is a critical step in the machine learning workflow. It measures the statistical relationship between variables, helping to identify which features have the most significant impact on the target variable, **Median House Value (MedHouseVal)**.

By analyzing the correlation coefficients, we can detect:

- **Positive Correlation:** Features that increase alongside house prices (e.g., higher income blocks).
- **Negative Correlation:** Features where an increase leads to a decrease in house value.
- **Multicollinearity:** High correlations between independent features that might affect model stability.

The following heatmap provides a visual representation of the correlation matrix for the California Housing dataset. Values closer to **1.0** represent a strong positive relationship, while values near **0** indicate no linear relationship.

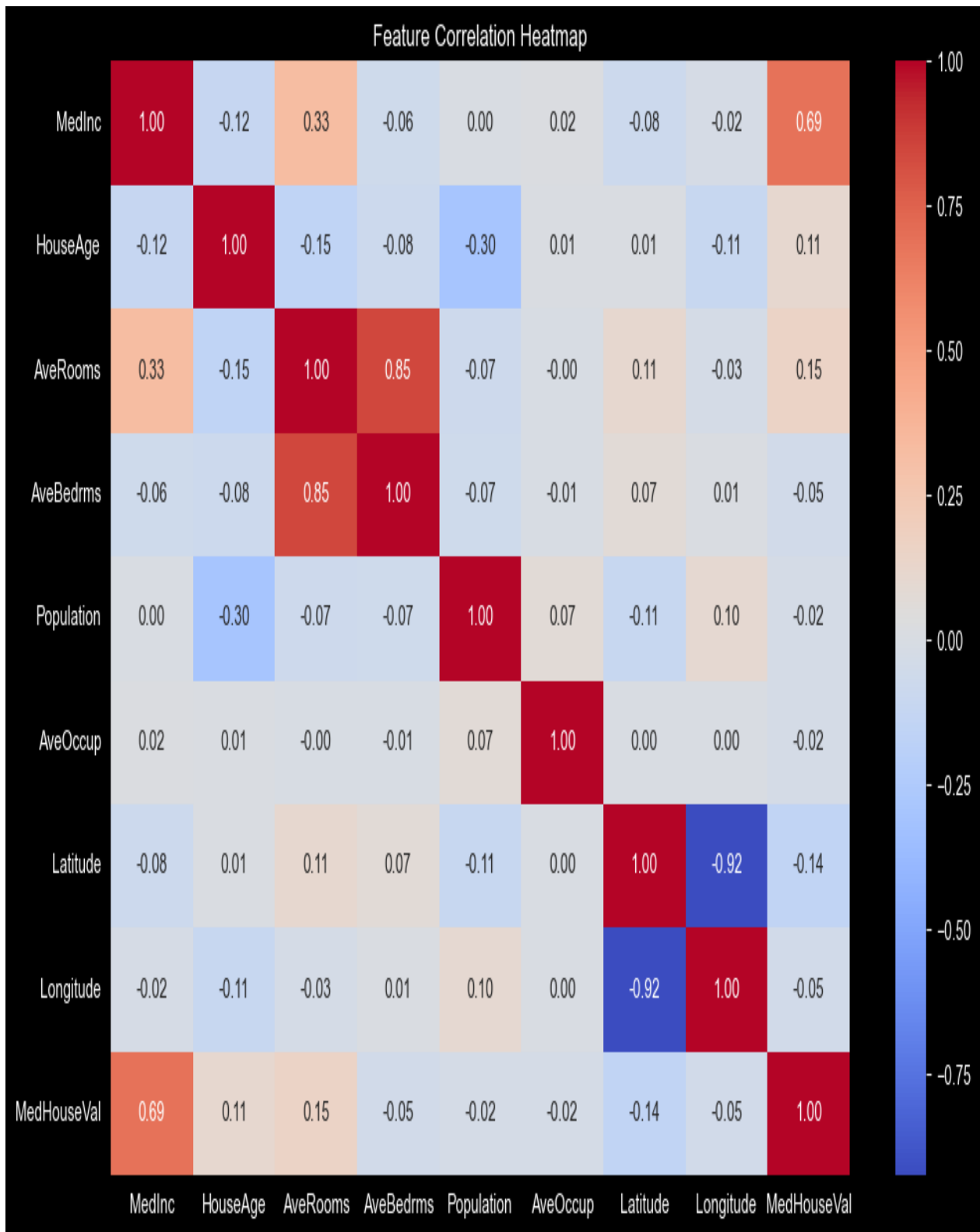


Figure: Feature Correlation Heatmap

Analysis:

The heatmap reveals that **MedInc (Median Income)** has the strongest positive correlation with the house price. This suggests that as the income of residents in a block increases, the property value tends to rise significantly. Other features like geographical location (Latitude/Longitude) show a secondary influence on pricing.

3. Model Development

3.1 Preprocessing

The data was split into a **Training Set (80%)** and a **Testing Set (20%)**. This ensures that the model is evaluated on "unseen" data to test its ability to generalize to new real-world inputs.

3.2 Algorithm

A **Linear Regression** model was chosen for this task. It works by finding the "Line of Best Fit" that minimizes the sum of squared differences between the actual and predicted values.

4. Results and Evaluation

To measure the success of the House Price Predictor, three key metrics were calculated:

- **Mean Absolute Error (MAE):** This represents the average magnitude of the errors in the predictions. A lower MAE indicates a more accurate model.
- **Root Mean Squared Error (RMSE):** This metric penalizes larger errors more heavily than MAE, giving us a sense of how much our outliers are affecting the model.
- **R² Score (Coefficient of Determination):**
 - **Interpretation:** The R² score tells us how much of the variance in house prices is explained by our model's features.
 - **Result:** [Insert your R² score here, e.g., 0.57]. This means that roughly 57% of the fluctuations in California house prices can be predicted using the features provided in this dataset.

4.1 Visualizing Predictions

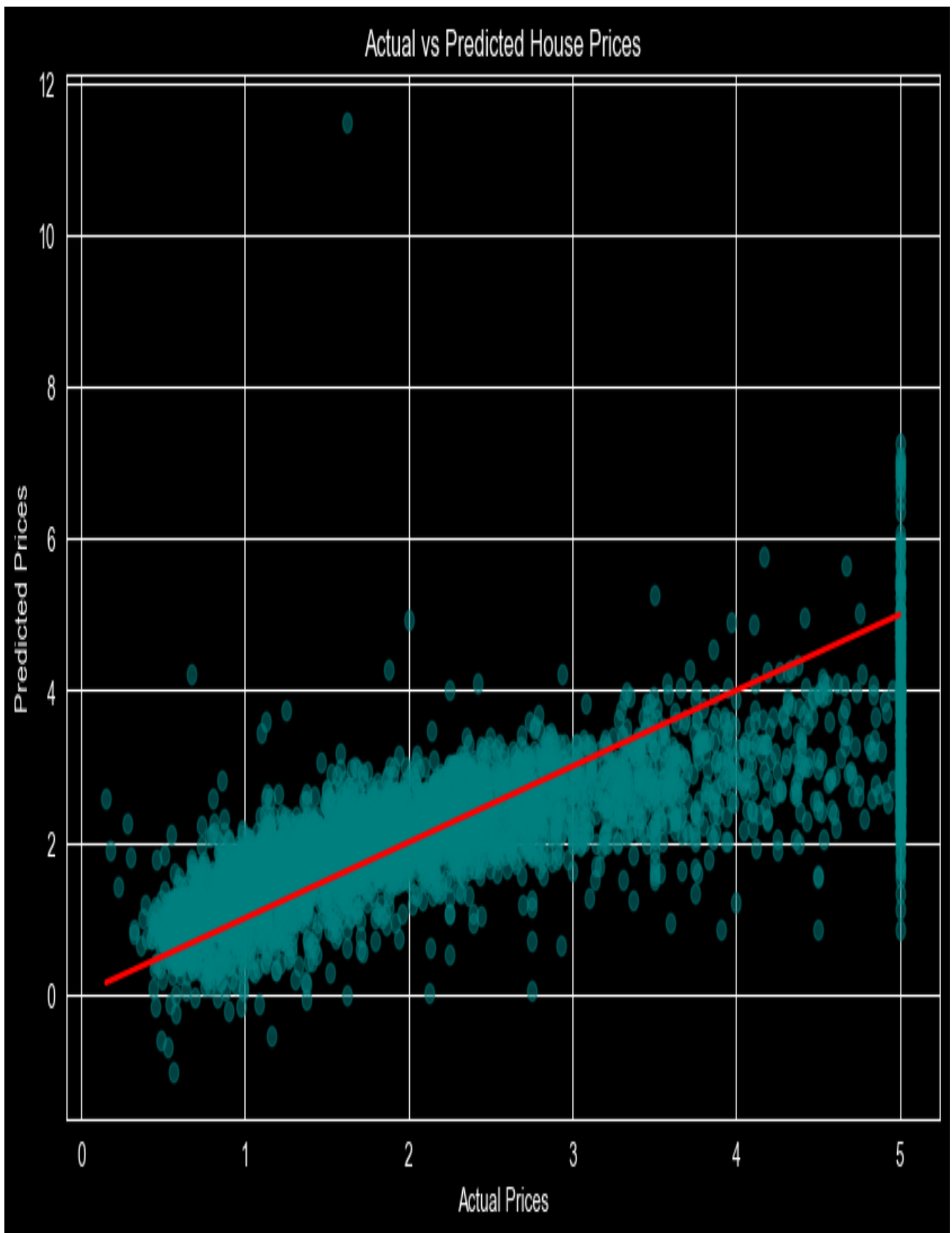


Figure: Actual vs. Predicted Scatter Plot

Observation:

The scatter plot shows that the predictions generally follow the diagonal line, indicating a strong relationship. However, there is some dispersion at the higher price points, suggesting that a simple linear model may struggle with very expensive luxury properties.

5. Conclusion and Future Improvements

The Linear Regression model provides a solid baseline for predicting house prices. While it captures the primary trend driven by median income, the model could be improved by:

1. **Feature Engineering:** Combining latitude and longitude to create a "distance to coast" feature.
2. **Advanced Algorithms:** Testing non-linear models like **Random Forest** or **XGBoost**, which might better handle the complex patterns in the California housing market.
3. **Handling Outliers:** Addressing the "price cap" seen in the dataset where values are capped at 5.0.